

Project Development Phase

Coding & Solution

Date	10/07/2026
Name	Karuru Arjun
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features implemented in the Voice-Based Concept Understanding Analyser (VBCUA) project. Each feature contributes to building an intelligent AI-powered educational assessment system that evaluates conceptual understanding and spoken communication skills using speech recognition, semantic similarity analysis, audio signal processing, and automated report generation.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Audio Upload	Upload MP3/WAV/M4A audio explanation.	Input Module	Done	High
2	Speech-to-Text	Convert speech into text using OpenAI Whisper.	Transcription Module	Done	High
3	Transcript Preprocessing	Clean and prepare transcript for semantic evaluation	NLP Module	Done	Medium
4	Semantic Similarity Analysis	Compare transcript with reference concept using Sentence-BERT	Semantic Analysis Module	Done	High
5	Audio Feature Extraction	Extract pause ratio, RMS energy, waveform and speech metrics.	Audio Processing Module	Done	High

6	Speech Fluency Evaluation	Detect filler words and evaluate communication fluency	AI Evaluation Module	Done	Medium
7	AI Evaluation Engine	Generate concept understanding score and qualitative feedback	AI Evaluation Module	Done	High
8	Interactive Dashboard	Display transcript, waveform, semantic score and analysis.	Streamlit Dashboard	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	10
Core / Must-Have Features	7
Additional / Nice-to-Have Features	3
Features Tested & Verified	10

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Audio Upload, Dashboard
2	Backend / Logic	5	Whisper, Sentence-BERT, AI Evaluation, Audio Analysis
3	Database / Storage	1	Audio Files, Reports, CSV
4	API / Integration	2	Semantic Similarity, Speech Evaluation
5	Security / Authentication	0	Not implemented in current version